

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think> To find the de Broglie wavelength of an electron with a kinetic energy of 1 eV , we need to follow these steps :

- 1 . Convert the kinetic energy from eV to joules .
- 2 . Use the equation for kinetic energy to find the velocity .
- 3 . Use the velocity and mass of the electron to find the de Broglie wavelength .

$1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$
 $\text{Kinetic Energy (KE)} = (1/2) * m * v^2$
where m is the mass of the electron ($9.109 \times 10^{-31}\text{ kg}$)
 $\text{KE} = 1.602 \times 10^{-19}\text{ J}$

So , $v = \sqrt{(2 * \text{KE}) / m)}$
 $= \sqrt{(2 * 1.602 \times 10^{-19}) / (9.109 \times 10^{-31})}$
 $\hat{=} 5.93 \times 10^6\text{ m/s}$

de Broglie wavelength ($\hat{=}$) $= h / (m * v)$
where h is Planck's constant ($6.626 \times 10^{-34}\text{ Js}$)

$\hat{=} (6.626 \times 10^{-34}) / (9.109 \times 10^{-31} * 5.93 \times 10^6)$
 $\hat{=} 1.23 \times 10^{-10}\text{ m}$ or $1.23 \times 10^{-7}\text{ cm}$ (because $1\text{ m} = 100\text{ cm}$)

Therefore , the de Broglie wavelength is $1.23 \times 10^{-7}\text{ cm}$.
</think>

<answer> $1.23 \times 10^{-7}\text{ cm}$ </answer>